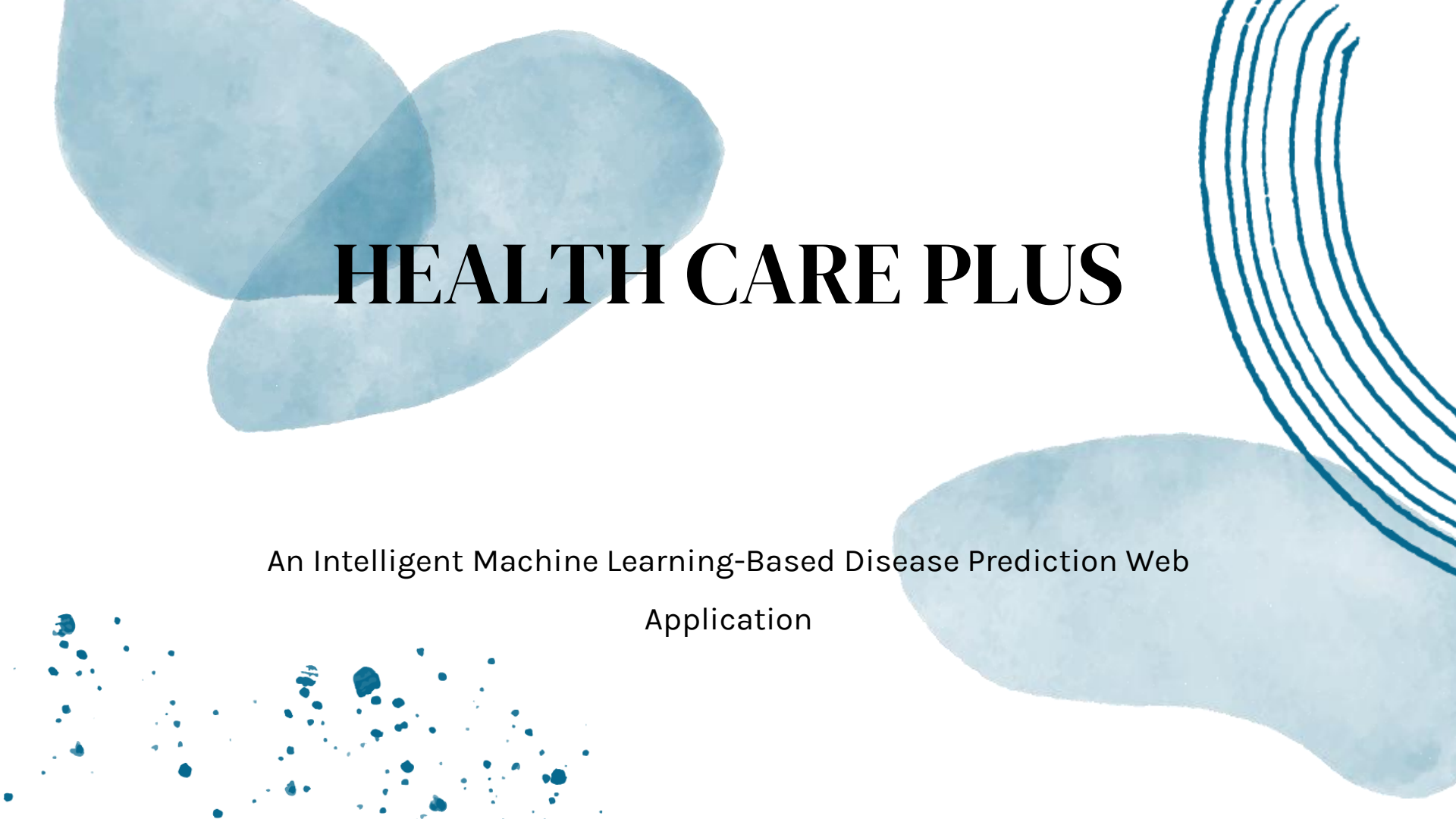


The background features several abstract blue watercolor-like shapes. In the top left, there are two overlapping, soft-edged blobs. On the right side, there are several thin, curved, parallel lines that sweep upwards. In the bottom left corner, there is a cluster of small, dark blue dots of varying sizes, resembling a splash or a group of particles. The overall aesthetic is clean and modern, with a focus on organic, fluid forms.

HEALTH CARE PLUS

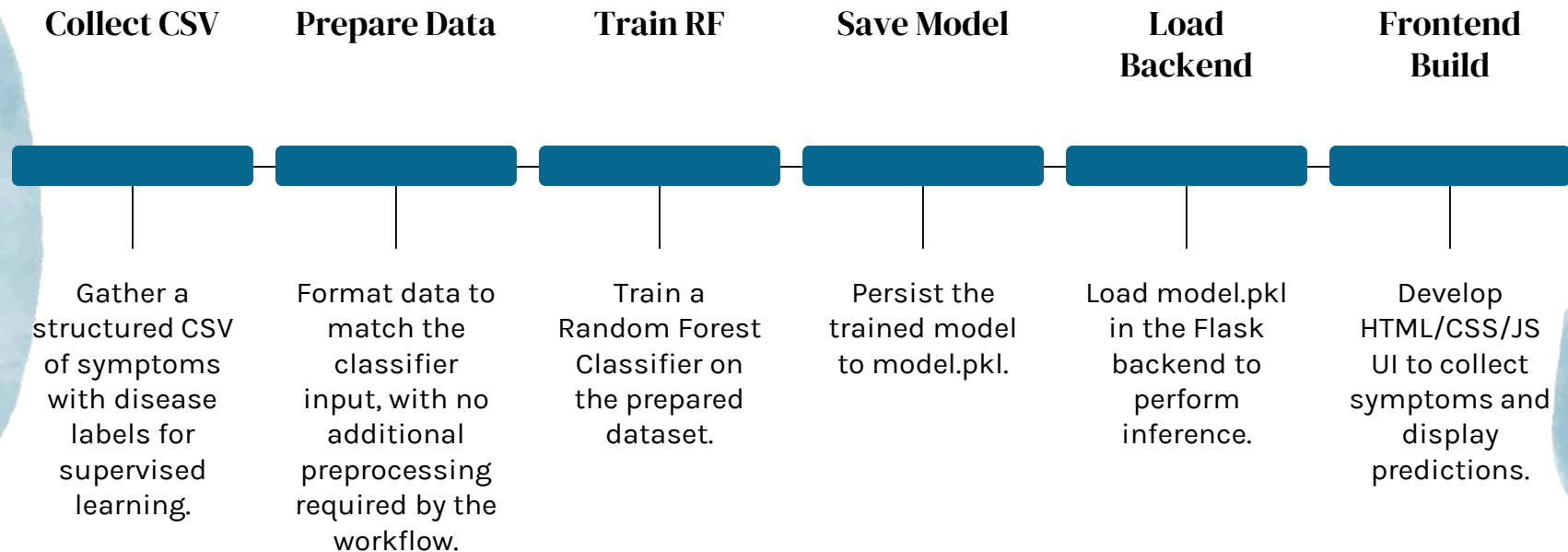
An Intelligent Machine Learning-Based Disease Prediction Web
Application

Care Access Gap



Many people cannot obtain timely clinical advice for common symptoms. They struggle to map what they feel to likely conditions, leading to delays and anxiety. This project reframes that as an access and interpretability problem. An interactive ML tool can offer immediate, relatable guidance to help users triage and seek appropriate care.

Implementation Timeline



End-to-End Flow

The application uses a **Random Forest Classifier** as its core Machine Learning model and **Flask** as the backend framework to provide an efficient connection between the user interface and the prediction system. Users can enter their symptoms through the interactive web interface, which are then sent to the Flask backend for processing. The backend passes the input to the trained Machine Learning model, where the Random Forest Classifier analyzes the provided symptoms and generates a possible disease prediction. The prediction is then returned by the backend and clearly displayed on the frontend, providing the user with an easy-to-understand result through a simple and interactive web application.

Tech Stack

- **Python** — Programming Language
- **Random Forest Classifier** — Machine Learning Model
- **Flask** — Backend Framework
- **HTML** — Frontend Structure
- **CSS** — UI Styling
- **JavaScript** — Frontend Interaction
- **CSV Dataset** — Data
- **model.pkl** — Trained ML Model
- **Web Application** — User Input & Prediction Display

Limitations & Next Steps

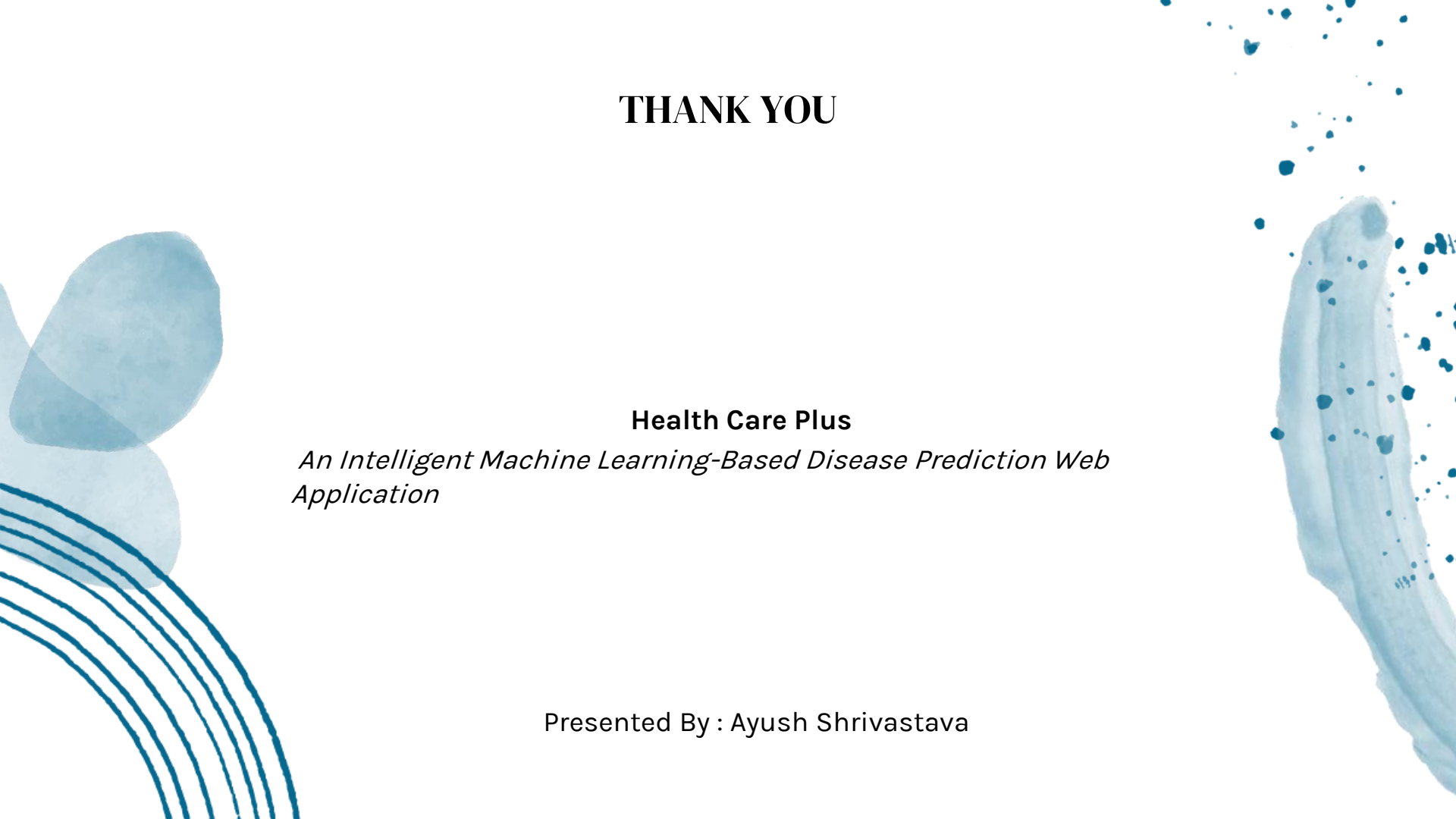
Limitations: no performance metrics or dataset provenance were provided; clinical validity and safety were not evaluated. Implications: without evaluation and explainability, results remain provisional for real-world use. Next steps: audit datasets, compute standard metrics, add model explainability (feature importance and local explanations), implement input validation and privacy safeguards, and

Feasibility Demonstrated

The prototype confirms rapid, symptom-driven disease prediction is feasible and provides a scalable platform for validation with clinical data. Gaps include missing performance metrics and unclear dataset provenance. Future work should add rigorous evaluation, data quality checks, explainability, and clinical validation to translate this approach toward practical use.

Conclusion

Health Care Plus successfully demonstrates the development of an end-to-end Machine Learning-based web application for possible disease prediction. By integrating a Random Forest Classifier with a Flask backend and an interactive HTML, CSS, and JavaScript frontend, the system provides a simple and user-friendly way to process symptom-related inputs and display a possible prediction. The project highlights the practical application of Artificial Intelligence, Machine Learning, API integration, and web development in a healthcare-oriented use case.



THANK YOU

Health Care Plus

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Presented By : Ayush Shrivastava